Management of two spotted spider mite, *Tetranychus urticae* Koch on tomato using phytoseiid predator, *Neoseiulus longispinosus* (Evans), at green house conditions

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Abstract

The two spotted spider mite, *Tetranychus urticae* Koch is a serious pest in greenhouses as well as on field grown crops. *Neoseiulus* (=*Amblyseius*) *longispinosus* (Evans) (Acari: Phytoseiidae) is a very effective predator for managing spider mites because of its feeding potential. The present study was undertaken to ascertain the efficacy of this predator, as a biological control agent for managing two spotted spider mite on tomato. Three weeks old tomato seedlings procured from commercial nursery were established in 22 cm diameter earthen pots using recommended practices, within a polycarbonate house, at GKVK, Bangalore. About 500 active stages of *T. urticae* were released on each 6 weeks old tomato plant (three weeks after transplanting) and allowed to feed for the next two weeks. During the 8th week, after estimating spider mite population in each plot, predators were released in four different prey: predator ratios: 50:1 (T2), 100: 1 (T3), 200: 1 (T4) and 400:1 (T5). Treatments were arraigned in a RCBD with four replicates. A group of plants maintained without predators served as control (T1). In T2 (prey : predator ratio 50: 1), predators required 5 weeks to eliminate the spider mites completely from the tomato plants, while at release ratio of 100: 1 (T3), predator eliminated the spider mites in 6 weeks. In T4 and T5 (predator: prey ratios 200: 1 and 400: 1), spider mites were not eliminated completely during the study period. Lowest proportion of eggs was found in T2 (0.74 eggs/leaflet) treatment followed by T3 (15.98), indicating predators prefer eggs than other stages of the spider mite, whereas most acaricides are not effective on eggs.

Keywords: Biological control, *Neoseiulus longispinosus*, Predatory mites, *Tetranychus urticae*

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Introduction

The two spotted spider mite, *Tetranychus urticae* is a serious pest in greenhouses as well as on field grown tomato. The mite generally feeds underneath the leaves and causes greying of the

leaves due to mesophyll collapse and yellowing. It is estimated that 18 – 22 cells are destroyed per minute by a single mite (Park & Lee, 2002). Continued feeding causes a stippled–bleached effect and later, the leaves turn yellow, grey or bronze. Complete defoliation may occur, if the mites are not controlled.

Predators are very important in regulating spider mite populations and should be protected whenever possible because of their great potential. Predatory mites can be distinguished from spider mites because of their long legs. The front pair of legs is often extended forward. They are more active and move faster than pest mites. The Phytoseiid predator, *Neoseiulus (= Amblyseius) longispinosus* (Evans) has a wide distribution and has ability to adapt to warm temperature inside poly-houses under South Indian conditions (Mallik *et al*., 1998).

Ganok (1982) reported that longevity of females of *A. longispinosus* was much higher than *T. truncatus* females. Kongchuensin *et al*. (2001) studied the effectiveness of *A. longispinosus* on *T. urticae* in a strawberry field in Thailand. When five predators per plant were released, when *T. urticae* number was five per leaflet; spider mite population was significantly reduced within four weeks after releasing the predator two times. Onkarappa (1999) and Mallik *et al*. (1998) reported that *A. longispinosus* caused maximum reduction of *T. urticae* population when they were released at a ratio of 1:300 compared to 1: 450 and 1:900 on rose plants. Sugeetha *et al.* (2007) reported that *N. longispinosus* had the ability to develop resistance to Monocrotophos and Dicofol.

Although tomato is an important vegetable crop grown in South Asia both for cooking and processing purpose and is often attacked by the two spotted spider mite, detail study on managing spider mites using predatory mites in tomato ecosystem is scant. The present study was undertaken to ascertain the efficacy of the phytoseiid predator, *Neoseiulus longispinosus* (Evans), as a biological control agent in managing the two spotted spider mite on tomato.

Materials & Method

This experiment was carried out in Polycarbonate house in Acarology section, Department of Entomology, UAS, GKVK, Bangalore, India during May - December, 2008.

Tomato plants, cv Abhinav (Syngenta India, Limited), were used in the present studies. About three weeks old seedlings procured from a commercial nursery were planted in 22 cm diameter earthen pots using recommended practices. The moisture level of pots was kept at field capacity using a drip irrigation system and temperature inside the glass house was maintained below 35^0C using blower-wet pad system during day time. Metal wire and jute threads were used to train tomato plants.

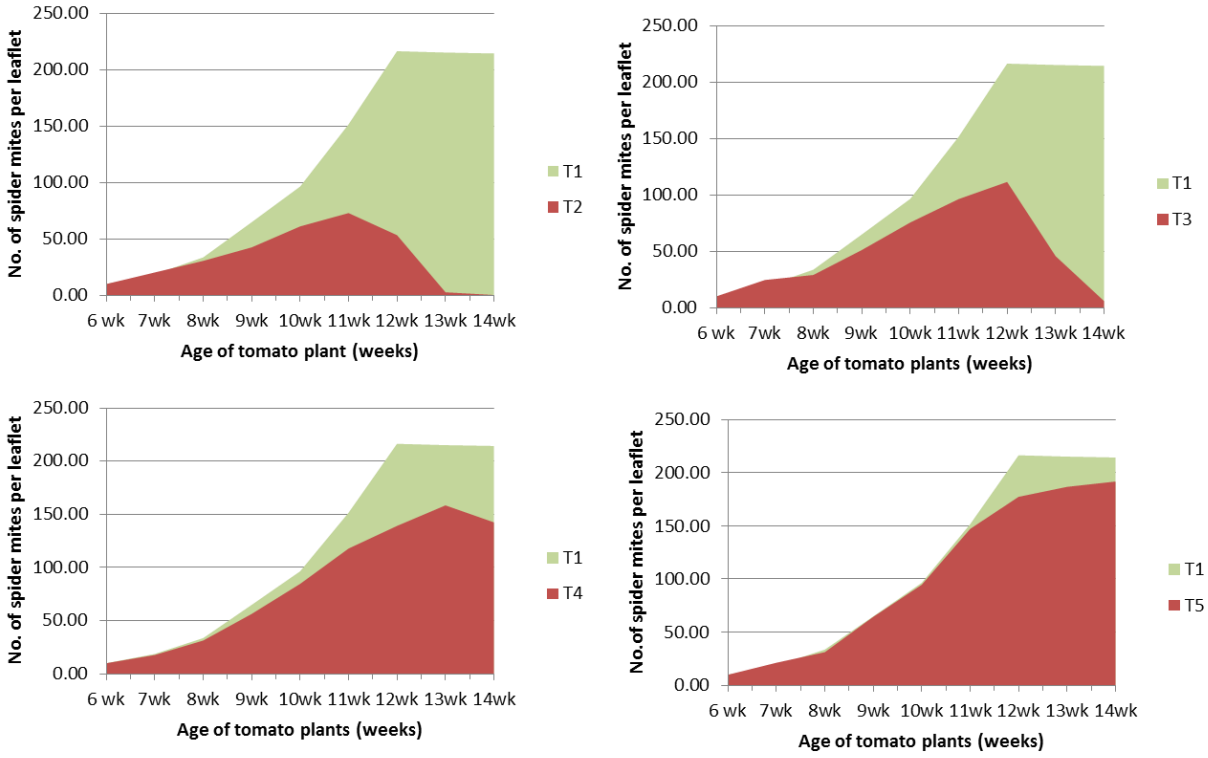
To evaluate the efficacy of *N. longispinosus* in suppressing population of *T. urticae* on tomato plants, about 500 active stages of the spider mite were released on each 6 weeks old tomato plant (three weeks after transplanting) in the experiment, and allowed to feed for the next two weeks. During the 8th week, after estimating spider mite population in each plot, predators were released in four different prey: predator ratios: 50:1 (T2), 100: 1 (T3), 200: 1 (T4) and 400:1 (T5). The predators were obtained from a laboratory culture reared with *T. urticae* prey mites on mulberry leaf bits. The treatments were replicated four times. A group of plants was maintained without predators served as control (T1). One tomato plant was planted in each pot and eight such pots were used for each treatment, they were arranged randomized complete block design. 160 pots were arranged in groups of eight, inside the polycarbonate house.

After the release of predators, observations were made every week, from 15 leaflets collected from five plants at three canopy levels (five each from top, middle and bottom canopies). The number of eggs and active stages of the spider mite and the phytoseiid predator on these leaves were counted under stereo-binocular microscope and the data collected up to 14 weeks, were used for studying efficacy of the predatory mite as a biological control agent. Analysis of variance (SAS Institute, 1989) and least significant difference (LSD) test at the 5% level was used to separate treatment mean values.

Results and Discussion

When the predators were released on spider mite infested tomato in different ratios, populations of the spider mite declined with time compared to the control plants (T1) on which no predators were released. On eight weeks old plants, before the release of the predators, spider mite populations were statistically similar in all the treatments. Significantly less number of spider mites was recorded in T2 (42.49 per leaflet) during 9th week, followed by T3 (51.09), T4 (56.34), T5 (64.44) and T1 (68.83). T1 and T5 were statistically on par. Similar trend was observed

during 10th and 11th week and spider mite population in T2 reached a peak (72.93 per leaflet) during 11th week, and then declined. During the 12, 13 and14 weeks, the number of spider mites in different treatments was significantly different from each other and it was highest in T1 (214.32 per leaflet) followed by T5 (191.72), T4 (142.53), T3 (5.09) and T2 (0.30) at 14th week. Spider mite population in T3 and T4 treatments reached peak (111.44, 158.48 per leaflet respectively) during the 12th and 13th week, respectively and then declined. Mite population in T5 treatment gradually increased till 14th week. Mite population in control (T1) reached the peak (216.36) between 12th – 14th weeks (Figure).



Spider mite population (eggs and active stages) when predators were released on 8 weeks old tomato plants (2 weeks after prey release)at different prey: predator ratios (T_1- zero predators, T_2- 50: 1, T_3- 100: 1, T_4-200:1 and T_5-400:1 of prey: predator ratios).

Prey: predator ratio of 50:1 (T2) was most effective, in which maximum control of the spider mites was observed compared to other ratios. Spider mite population increased up to 3rd week of predator release (11 weeks old plants) from 30.55 to 72.93 per leaflet and started to decline at T2. Population of the prey mite in T2 treatment was significantly lower than in control plots and in other treatments from 9 to 14 weeks after planting, during the 14th week the total spider mite

population was 0.3 per leaflet. Significantly less number of spider mites were observed in T3 treatment (100:1 prey: predator ratio) than in T4 (200:1 prey: predator ratio), T5 (400: 1 prey: predator ratio) and T1 (control). Populations of eggs and active stages of the spider mite changed in a similar pattern.

Since the performance of a biological control agent is affected by several factors, their initial numbers being one, the appropriate prey: predator ratio should be determined to decide the number of predators to be released in a cropped area. The results from this study indicated that when the release ratio was 50: 1 (T2), predators required 5 weeks to eliminate the spider mites completely from the tomato plants, while with release ratio of 100: 1 (T3), predator eliminated the spider mites in 6 weeks. When release predator ratios were 200: 1 and 400: 1, spider mites were not eliminated completely during the study period. Hence for *N. longispinosus* the effective ratio might be 25: 1 (prey to predator) to suppress *T. urticae* within a short period of time before economic damage is caused to the plants by the pest.

In rose, complete elimination spider mites were recorded within 3 weeks (Onkarappa, 1999 and Mallik *et al*., 1998) at the ratio of 1: 300 predator: prey, but same was not observed even at 1: 50 predator: prey ratio in tomato. It is possible that the different types of glandular trichomes located on tomato leaves (Maluf *et al*., 2007), but not in rose leaves, or some volatiles emitted from tomato plants, as also the plant architecture, reduce the efficacy of predators, which needs to be ascertained. It is suggested that prey-predator interactions vary with plant to plant.

Proportions of different life stages (eggs and active stages) of the spider mite were calculated in different treatments of predator release. Significant lowest proportion of eggs was found in T2 (0.17^{d} per leaflet) treatment than T3 (0.29^{c}) T4 (0.38^{b}), T5 (0.37^{b}) and T1 (0.42^{a}) at 14th week. It was cleared that reduction of eggs of spider mites more under high predator pressures and this clearly suggested that predators prefer eggs than other stages of the spider mite. Most insecticides are not effective on eggs. Therefore, two or more applications of the miticide will be required at five-day intervals during the summer or seven-day intervals during the winter (Fasula *et. al*, 2008). So use of predatory mites to suppress the spider mite is very efficient alternative.

Reference

Fasula TR, Mizell RF and Shorl DEC 2008 Wood Bug. University of Florida. http://nfrec.ifas.ufl.edu/MizellRF/WoodyBug/index.html

Ganok V 1982 Preliminary study on the biology of cassava red mite, *Tetranychus trancatus* Ehara and the predacious mite, *Amblyseius longispinosus* (Evans). *Mag. Sc. Thesis*, Kasetsart University, Bangkok, Thailand. 79 p.

Kongchuensin M, Charanasri V, Saringkapibul C and Kulpiyawat T 1989 Biology of the two-spotted spider mite, *Tetranychus urticae* Koch and its predatory mite, *Amblyseius longispinosus* (Evans) on strawberry. *Entomol. and Zool. Gazette*, **11**: 195–204.

Mallik B, Onkarappa S, and Kumar HM, 1998 Management of spider mites, *Tetranuchus urticae* Koch on rose using phytiseiid predators, *Amblyseius longispinosus* (Evans) in polyhouse. *Pest Mangt. Hort. Ecosyst*., **4** (1): 46 -48.

Maluf WR, Inoue IF, Ferreira RPD, Gomes LAA, De Castro EM and Cardoso MG 2007 Higher glandular trichome density in tomato leaflets and repellence to spider mites. *Pesq. Agropec. Bras., Brasília*, **42** (9): 1227-1235.

Onkarappa S 1999 Management of two spotted spider mite, *Tetranychus Urticae* Koch. (Acari: Tetranychidae) on rose. *Ph.D Thesis*. Department of Entomology, University of Agricultural Sciences, Bangalore. 190 – 195 pp.

Park YL and Lee JH 2002 Leaf cell and tissue damage of cucumber caused by two spotted spider mite (Acari: Tetranychidae). *J. Econ. Entomol*. **95:** 952–957.

SAS Institute 1989 Version 6.12. SAS institute Inc., Cary NC.

Sugeetha G, Mallik B and Sannaveerappanavar VT 2007 Development of insecticide resistant strains of *Neoseiulus longispinosus* (Evans) (Acari: Phytoseiidae). *J. Acarol*., **17** (1& 2): 61 - 62.